PAPER_671: UQFF dM/dt Derivation

Subtitle: Full step-by-step derivation of the UQFF-modified mass loss rate, with analytic M(t) approximation and Euler simulation. **Module:** UQFFDMDtDerivation
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Abstract

We present the complete three-step UQFF derivation of the modified black hole mass loss rate dM/dt, showing a ~33× suppression relative to standard Hawking emission.

1. Standard Hawking Rate

$$\left.\frac{dM}{dt}\right|_{std} = -\frac{L_H}{c^2} = -\frac{\hbar c^4}{15360\pi G^2 M^2 c^2}$$

2. UQFF Step 1 — f_TRZ

$$\frac{dM}{dt}' = \left.\frac{dM}{dt}\right|_{std} \cdot (1 - f_{TRZ})$$

Physical basis: negentropic restoring force suppresses pair production at horizon.

3. UQFF Step 2 — [UA]/[SCm]

$$\frac{dM}{dt}'' = \frac{dM}{dt}' \cdot \frac{\rho_{SCm}}{\rho_{UA}}$$

4. UQFF Step 3 — U_m Anchor

$$\left.\frac{dM}{dt}\right|_{UQFF} = \frac{dM}{dt}'' \cdot \exp\left(-\frac{U_m}{k_B T_H}\right)$$

5. Combined Formula

$$\left.\frac{dM}{dt}\right|_{UQFF} = \left.\frac{dM}{dt}\right|_{std} \cdot (1 - f_{TRZ}) \cdot \frac{\rho_{SCm}}{\rho_{UA}} \cdot \exp\left(-\frac{U_m}{k_B T_H}\right)$$

Suppression factor: $0.9 \times 0.1 \times e^{-1} \approx$ **0.033** (evaporation ~30× slower).

6. Analytic Approximation

$$M(t) \approx \left(M_0^3 - 3A\,t\right)^{1/3}$$

where A = |dM/dt|_UQFF × M_0^2.

7. C++ Module

`UQFFDMDtDerivation.h / .cpp` — Session 172 CP4 #255 — `UQFFDMDtDerivationCalculator`

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